

Supplementary Material

Automatic Detection of Morphological Anomalies in Altiplano Alpacas Using a Computer Vision System: A Curated Dataset, Compact Detector, and Ocular Anomaly Classification Feasibility Study

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These figures provide detailed diagnostics for the ocular-anomaly-classifier feasibility study (Section 3.2 of the main text). Because the leakage-free classifier performs at chance ($\text{AUC-ROC} = 0.506$, 95% CI $[0.366, 0.645]$), they are reported here as supporting evidence of the negative result rather than in the main text.

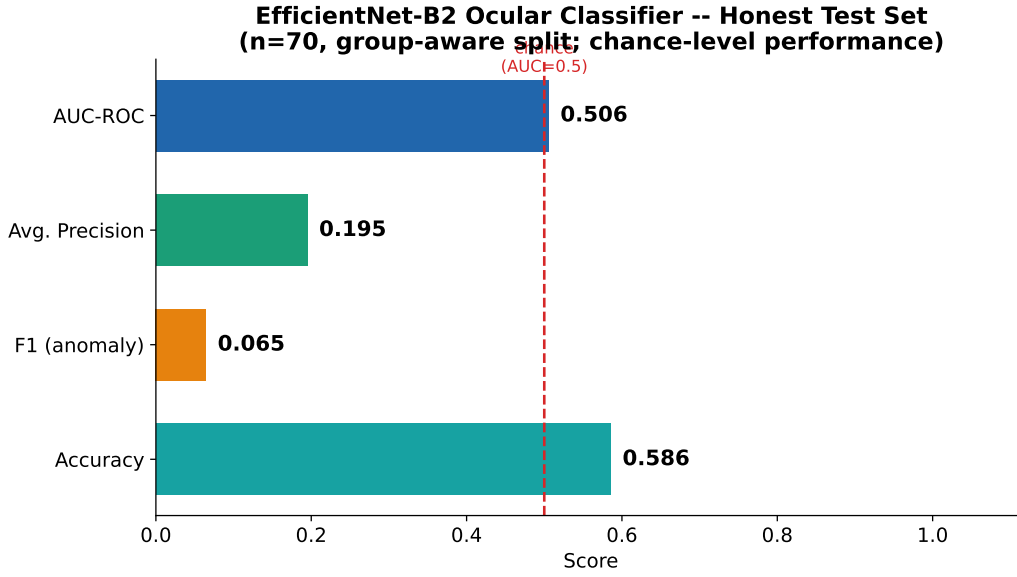


Figure S1: EfficientNet-B2 ocular anomaly classifier: leakage-free test-set metrics ($n = 70$, group-aware split). $\text{AUC-ROC} = 0.506$ sits on the chance line (dashed), with Average Precision = 0.195, $\text{F1 (anomaly)} = 0.065$ and $\text{Accuracy} = 0.586$.

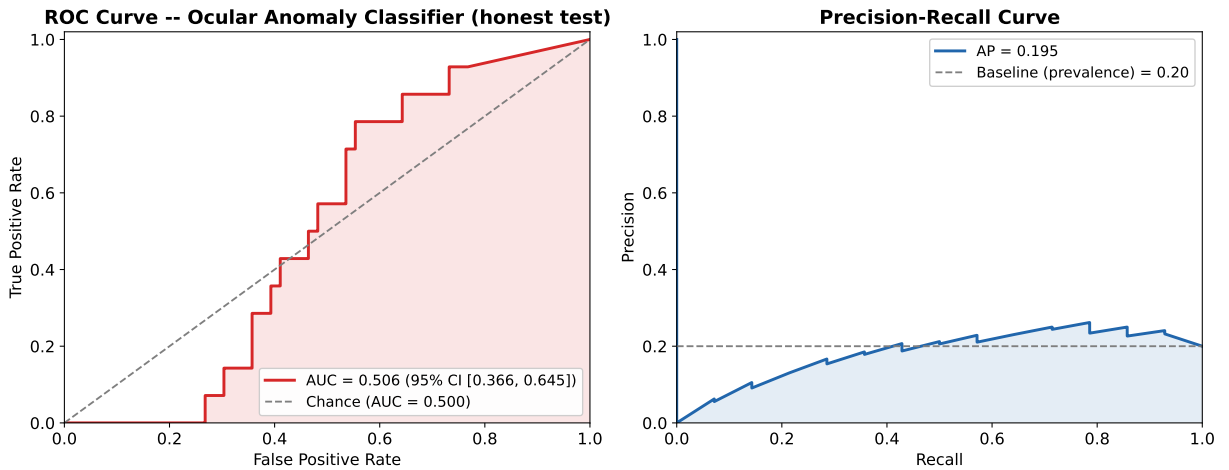


Figure S2: ROC and precision–recall curves for the classifier under the leakage-free protocol (bootstrap $B = 1000$, seed = 42). The ROC lies on the chance diagonal ($\text{AUC} = 0.506$); the PR curve tracks the prevalence baseline ($\text{AP} = 0.195$).

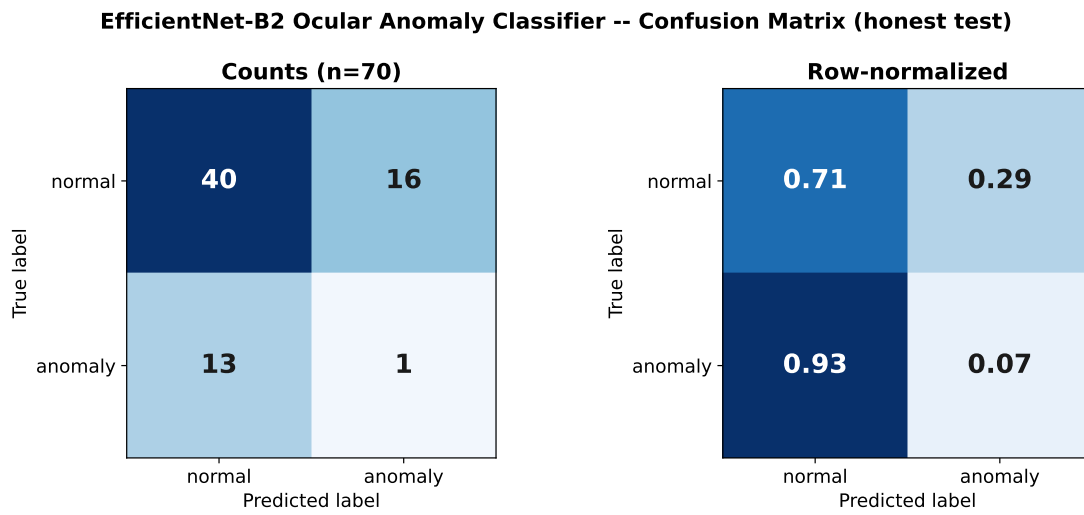


Figure S3: Confusion matrices for the classifier: raw counts (left, $n = 70$) and row-normalised (right). Of 14 true anomalies only 1 is correctly identified, consistent with chance-level discrimination.